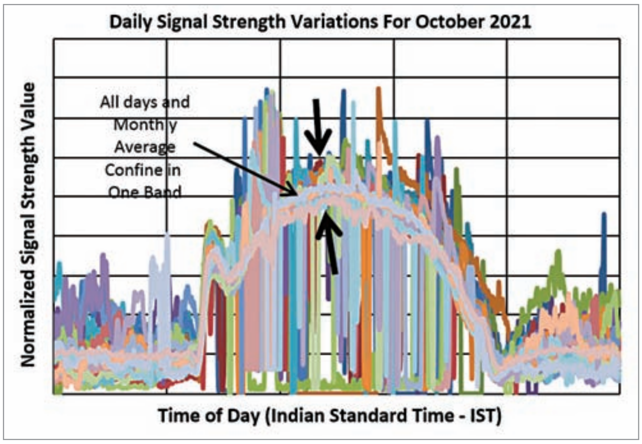

Fig. 5(a): The all-days plot with monthly average for October 2021. Notice all days in a band

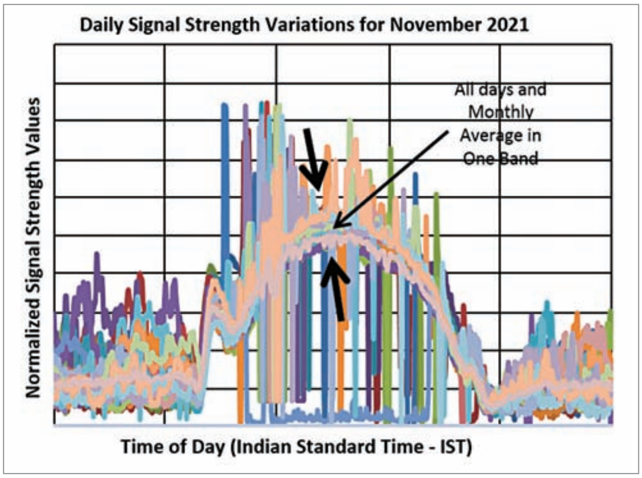

Fig. 5(b): The all-days plot with monthly average for November 2021. Notice the single band

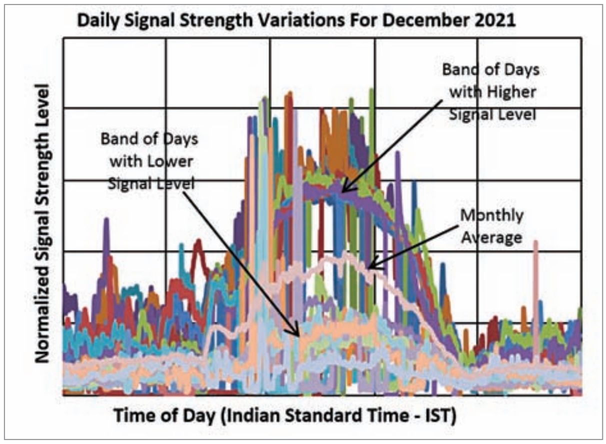

Fig. 5(c): The all-days plot with monthly average for December 2021. Note the split in days above and below average line

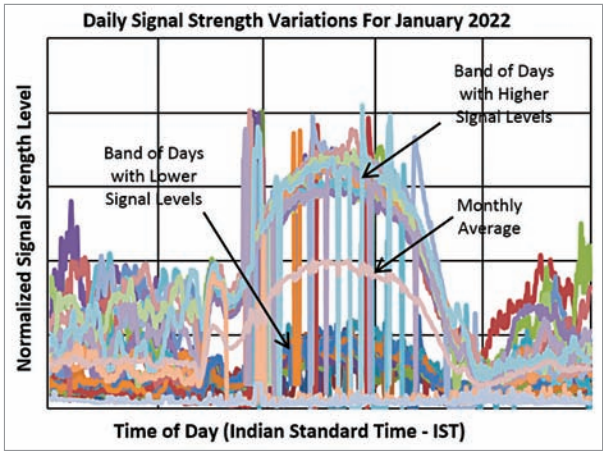

Fig. 5(d): The all-days plot with monthly average for January 2022. Note the very well demarked appearance of days above and below the average value

in a systematic manner the experimental setup, data observed, and the findings after simple analysis of the data.

Experimental setup and data observational method

The experimental setup consists of VLF receiver tuned to 18.200kHz frequency. A square loop antenna of 1m × 1m picks up the radio signal at this frequency and feeds to the receiver.

Fig. 2 depicts the block diagram of the receiving setup. The output from multiple VLF receivers tuned to different frequencies is fed to a data logger. The transmitter-receiver combination forms a VLF path in that specific direction. Receiving signals simultaneously with different frequencies and from different directions is therefore called the Multi-Path-Multi-Frequency VLF technique.

Each of the receivers actually acts as a tuned RF level indicator that gives a DC voltage proportional to the signal strength of the input RF/VLF signal from the antenna. This output voltage ranges from 0-10V DC for a signal input from -115dBm to about -55dBm with a noise floor at around -120dBm for each of these receivers. All receivers conform to same calibration curve for the input signal and output DC levels.

This output is then fed to a standalone datalogger with on-board storage on a memory card. At the same time, it relays data digitally to a local PC for real-time display purposes.

For the present case, the signal from VLF station VTX3 operated from India is monitored and the transmitter-receiver VLF path and the associated seismic and volcanic sensing zones are shown in the map in Fig. 3.

This setup runs and records data 24 × 7 × 365, except during power outages. As such, the data records the signal strength variations of each VLF channel as a function of the time of the day. For data analysis, data is sorted day-wise and collated month-wise. An average for each month is calculated and each day of that month is then compared